

Do Quality, Environmental, and Social (QES) Certifications Improve International Trade? A Comparative Grey Relation Analysis of Developing vs. Developed Countries

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ABSTRACT

The purpose of this empirical study is to undertake a comparative analysis of developing vs. developed countries based on adoption of ISO 9001, ISO 14001 and SA8000 certifications and their impact on international trade. We utilize a combination of models including Deng's Incidence Analysis (GIA), Absolute Degree GIA, Second Synthetic Degree GIA (SSDGIA) and Conservative (maximin) approach to inform decision making under uncertainty. Data was collected from the ISO and SAI official websites of the top fifteen certified countries for the period of 2002 to 2017. Additionally, export of goods and services data was gathered from the World Bank. The results reveal that the adoption of Quality, Environment and Social (QES) standards have a positive and significant effect on exports of goods and services in developing countries. While this is interesting, we also find that ISO 14001 certification contributes more to economic development than ISO 9001 and SA8000 in both developed and developing countries, with developing countries showed superior performance. This study is the first of its kind in evaluating the association of QES standards on international trade in developed and developing countries while using multiple grey relational analysis models. Finally, this article proposes recommendations for policymakers to improve international trade and sustainable development.

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1. Introduction

The importance of international trade and economic development are closely intertwined. Many researchers have accepted the idea that international trade enhances economic growth of a country Sokolov et al. [1], Omri et al. [2]. It is reported that international trade supports the economies of developing countries through the upgrading and improvement in industrial infrastructures, providing facilities to land capital area, and by helping development through technological progress and innovative resources for institutional advancement Lee and McKibbin [3]. Building on this context, economic

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development increases the effect of international market competition and also expands upon its efficiency Wagner [4]. Therefore, international and domestic companies are facing more challenges and have become more competitive resulting in higher production and enhanced levels of domestic products. It is also noticed that with the implementation of new progressive developments, domestic firms have become more developed and are providing innovative products that are exported to developed countries Harrison and Rodríguez [5]. It is also noted that new and emergent technologies have changed the trajectory of domestic businesses and have enabled them to be more developed and efficient Marquis and Raynard [6]. Therefore, international trade has as significant impact on improving the lifestyle standards and progressive actions taken by domestic firms the developing countries.

In 1987, the International Organization for Standardization (ISO) introduced Quality Management System (QMS) ISO 9000 certification standards for products, frameworks, and administrative systems so that organizations can fully determine their dimensional values through the structure of dominant power in quality support environment. The goal was to enable various associations to ensure that they meet the quality standards and report the issues of customers and different market competitors in order to strengthen administrative requirements and satisfaction levels that are examined reliably with productive item or administration services Kakouris and Sfakkianaki [7].

In order to achieve the ISO 9001 standardization certificate, organizations must pursue requirements found within the framework of the ISO 9001 standardized rules Psomas and Pantouvakis [8]. Achieving the popularity and widest scope of ISO 9001 certification, the quality standard has the capability and necessary fundamentals to allow productive requirements under desired specifications, and thus, it generates a positive effect on trade. Many research studies have examined the impact of ISO 9001 and the standardized adoption of effective parameters that help to introduce its effect on international trade of developed and developing countries Dellana and Kros [9], Martincus [10]. The findings of the influence of management standards on exports is complex. With the advanced developmental growth in the international trade sector, various developing countries are adopting the certification standards of ISO 9001, Environmental Management System (EMS) ISO 14001, and (Social Accountability) SA 8000 Daddi et al. [11], Heras et al. [12]. The basic requirements of these international trade system have grown in developing countries to improve the processes for quality outcomes and productive achievement. Building on prior research involving these standards and certification procedures, we want to better understand the interactions of these standards, country context, and levels of performance. To this end, we have collected data across countries, review the literature and later analyze this data in order to draw valid conclusions on the basis of the influences of ISO 9001, ISO 14001, and SA 8000 (QES). We are particularly interested in learning more about international quality standards relationships to exports in the context of developed and developing countries Cao and Prakash, [13], Castka [14].

Challenges to the adoption of ISO certification standards are complex in the matters of universal trade which has left some organizations behind in developing countries. Among the various reasons why this is happening include developed nations, when dealing with the complexities of a worldwide economy, assert a need for quality products and use of standards to help verify this [15]. Looking deeper into the structure of the ISO 9001 standard, the certification procedure has been developed with the aim to promote and make improvements in the global exchange of goods. The research within this paper aims to provide an expanded strategy for quality grounded in observational research by assessing the impact of international management standards on exports among developing and developed nations.

According to ISO [16], the ISO certification standards are considered to be voluntary and collaboratively developed. Enrollment in these certifications have 164 bodies that are federally administered and possesses influence and impact regarding the guidelines being mentioned in governmental rules in developed countries such as, U.S and abroad Salim et al. [17]. These certification standards are designed to be reliable to business managers and represent a shift in global trends regarding quality and sustainability Carnerud [18]. An organization can establish ISO 9001's, ISO 14001's and SA 8000's essential prerequisites while utilization pre-designed models and frameworks for preparing and recording documentation McAdam et al. [19]. The development and evolution of these standards has not gone unnoticed by business decision makers and researchers.

An abundance of past research has shed light on the scope and importance of ISO 14001 for environmental management that has gained wide attention in both developed and developing countries Ikram et al. [20]. Policy makers and organizations alike appear to find guidance in the standards and consider them as a basic component for mutual cooperation between regulators and organizations. Studies also highlight that many developing countries are now effectively seeking avenues of compliance with the implementation of ISO 14001's goals and strategies within underlined its framework. For example, Egypt has been concerned with the practical applications of the broadest certification management system of ISO 14001 Tayo et al. [21]. The majority of the successful firms in the European Community (EC) have focused on the instructions and recommendations as given by the most prevalent European organizations to cope up with business processes and management systems through use of ISO 14001.

The objective of the current study is to empirically investigate the impact of ISO 9001, ISO 14001, and SA 8000 certifications on the export of goods and services for the top fifteen certified developed and developed countries, based on available data from SAI [22] and ISO [16] for the period 2002–2017. Advanced mathematical decision making models have been used to empirically investigate relationships, namely grey relational models i.e., Deng degree of grey relation and absolute degree of grey relation models. Additionally, we employed Synthetic Degree of Grey Incidence Analysis (SSDGA) models as proposed by Javed and Liu [23] and was used to analyze the satisfaction among public and private sectors. Contributions of this study will be to increase the understanding of the expected ISO 9001, ISO 14001, and SA8000

certification performance regarding international trade in developed and developing countries. The study investigates the association of management standard adoption on country exports of goods and services as compared to developed countries. Finally, the study provides valuable information for decision makers in while assessing an international comparison of countries within a dynamic decision making environment.

This paper is organized as follows: Section 1 is the introduction; Section 2 presents the background/literature review of Quality Management System ISO 9001, Environmental Management System (EMS) ISO 14001, Social Accountability SA8000 and the current scenario in developed and developing countries; Section 3 describes the methods used to check the association of certified countries on their economic position in the given scenario; Section 4 presents the results and discussion. Finally, Section 5 concludes the study, with limitations and future research direction.

2. Literature review

Previous research studies Zahonogo [24], Vogiatzoglou [25], have found that developing nations are experiencing strong progress in international trade which has led to positive impacts on their economic resources. With the rapid expansion and modernization of emergent industries in the developing world, decision makers have been looking for modern business operating tools and to adopt verified testing quality standards. To this end, ISO 9001 has improved the quality of products prepared by ISO certified organizations Valmohammadi and Kalantari [26].

ISO 9001 certification is seen by companies as a way to influence their export performance, by way of decreasing data gaps and uncovering internal opportunities for improvement. Surprisingly, regardless of the wide ranging dissemination of this standard, little is noted about the effect of its adoption on a company's export performance. Most of the studies related to ISO accreditation and companies' exports includes only descriptive information. For instance, Castagnino [10] depicts a company's export patterns in relation to its accreditation status and size in Argentina. In this case, the exports of small and medium size companies with ISO accreditations developed more than those of peers without accreditation and those of high profile exporters, yet the authors pay little attention to whether or not companies had utilized ISO.

The registration of ISO 9001 is found to be an initial and vital step of the verification procedure for continuous process improvement that provides association with sustainability improvement and learning practices within the boundary of a whole organization Manders et al. [27]. It is also important to note that there is little information at the industry level, particularly in developing nations Leonidou et al. [28]. In one study, Lefebvre and Lefebvre [29] utilized a sample of 3032 Canadian manufacturing companies and observed that the adoption of ISO 9001 was related to both an increased likelihood of exporting and the share of exports as a percentage of total sales, particularly in high-innovation enterprises. The impact on trade regarding the adoption of ISO 9000 fundamentally centers on accreditation as a device for exporters to signal quality to purchasers and to decrease data asymmetries and transaction costs. ISO accreditation is a reliable tool with which clients can recognize high quality production Cao and Prakash [13]. Finally, the adoption of international management standards have positive economic impacts which are exhibited when the reduction in transaction cost exceeds the accreditation cost Sroufe and Curkovic [30].

According to ISO (2018), nearly one million organizations and trade associations in over 170 nations have confirmed adoption of the ISO 9001 certification. ISO 9001 builds transparency in quality management system that help to provide developmental strategies and consistently verify processes. This can help improve the quality reputation of business so that suppliers can identify directions for future development Sroufe and Curkovic [31]. Basically, the effect of ISO 9001 accreditation has fundamentally strengthened exports all over the world.

The United Nations Industrial Development Organization has reported that ISO 9001 certification has satisfactory measures to observe outcomes, enhances customer satisfaction and increases the exports of developing countries UNIDO [32]. Since 2011, China has been considered to be an important country for extending ISO 9001 norms as compared to other nations by overtaking nearly 40% of the top 10 developed countries Heras-Saizarbitoria and Boiral [33]. Consequently, ISO and Social Accountability standards provide a new type of opportunity to developing countries and the companies making products within them.

Adoption of the ISO 14001 standard encourages organizations to be more environmentally conscious in their processes internally, and export business strategies externally in a variety of ways. First, it encourages companies to increase transparency regarding their sustainable practices to different interested parties, e.g., government authorities, investors, and residents Castka and Prajogo [14]. Second, it enables companies to exhibit a sustainable corporate image in order to attract clients who are environmentally focused. It also assists companies in maintaining a strategic distance from negative consequences, for example, paying violation penalties, facing government interference, and clients boycotts De Oliveira et al. [34]. ISO 14001 enrollment entices other firms to proceed with the ISO procedure Ozusaglam et al. [35]. Finally, rising nations are commonly acquainted with the idea of voluntarily implementation if EMS practices Singh et al. [36]. In the world market, an organization needs to showing good business practices to clients and those within rising nation markets also look to ISO 14001 accreditation to signal better practices than businesses without this Berényi [37]. Numerous countries have come to incorporate an EMS, such as ISO 14001. When appropriately regulated and connected these standards provide a mechanism that can improve environmental performance for organizations of differing sizes in order to fulfill customers' demands [38]. In nations where requirements and abilities are not well established, governments have a propensity for well-established standards such as ISO 14001 [39].

Research assessing ISO 14001 certifications has demonstrated improved environmental performance Sroufe [40] and it has impact on international trade Prakash and Potoski [41]. ISO 14001 equips organizations with explicit rules and

methods to execute and improve an EMS. In an ongoing global review directed by Summers [42], the advantages of the ISO 14001 accreditation were evaluated in 15 developed and developing nations. They observed that most organizations were satisfied with ISO 14001's effect on their organization's benefits and capacity to trade internationally.

Building on the success of ISO standards, the SA8000 standard is the primary social standard worldwide which may be utilized for third party accreditation. It enhances and protects the privileges of employees in an organization through socially responsible activities and supports reasonable and fair working status. Wang [43] examined the SA8000 standard as another sort of trade barriers that influences the competitive benefits of China's export products and restricts the amount and size of export products. In a recent study, organizations acknowledged that the adoption of ISO 14001's effect on their organizations was beneficial with positive relationships to profit and international trade Liston and Heyes [44], and we predict similar relationships with SA8000.

The adoption of international standards such as SA 8000 has a significant role in growth of international trade. A study conducted by Ciliberti et al. [45] investigated that SA8000 certification adoption and its level are connected with increase in countries bilateral exports of goods and services, specifically in developing countries scenario, SA8000 certification adoption helps out to tackle the social challenges as well as enhance company exports Christmann and Taylor [46]. Similarly Crals and Vereeck [47] observed the effect of SA8000 standard diffusion on international trade. They investigated that SA8000 certification has a push or pull effect on exports of goods and services. The authors investigated that SA8000 certification have significant positive effect on trade in the exporting countries but no or negative relationship in the host countries Luken and Stares [48].

3. Methodology

This section discusses the approach used within this study for our data collection, along with the grey models used for analysis.

3.1. Data collection and analysis

In this study, we empirically assess the association of ISO 9001, ISO14001 and SA8000 for its relationships to performance and international trade. To start, the top ISO9001, ISO 14001 and SA8000 certified countries were selected from around the world. The countries included are: China, Italy, Japan, Spain, Germany, USA, India, France, Brazil, UK, Romania, Pakistan, Bulgaria, Vietnam, and Portugal. The ISO 9001, ISO 14001 and SA8000 data used in this study is from the period 2002–2017 and was gathered from the International Organization for Standardization (ISO) and Social Accountability International (SAI) official websites. We also included the exports of goods and services as measured in US dollars, using data collected from the fifteen selected countries from the World Bank official website from 2002 to 2017. To enable our analysis with the application of multiple models, Nanjing University of Aeronautics and Astronautics (NUAA)'s Grey System software (v8.0) was used to gauge the impact of these standards on the international trade of developed and developing countries by using advanced mathematical decision making tools, namely Deng's Grey Incidence Analysis (GIA) model, Absolute Degree GIA model, and a new proposed Second Synthetic Degree GIA (SSDGIA) model. Further, the Conservative (maximin) model was employed for decision making under uncertainty. The SSDGIA model was designed with the use of MATLAB and MS Excel.

3.2. Grey Relational Analysis (GRA) models

The popularity of Grey System Theory, particularly GRA models has increased since last two decades. The application of GRA models has been employed in numerous fields such as management, engineering, medical, environmental sciences, energy and others Senthilkumar and Muralikannan [49], Wang et al. [50], Arce et al. [51], Yazdani et al. [52]. The GRA model provides best optimal solution to conventional and famous approaches of data analysis like statistics. The superiority of GRA models to make decision on small size of sample. Firstly, we normalized the certifications and exports data of top sixteen countries from 2002–2017 for analysis. Secondly, GRA models was utilized to check the association of ISO 9001, ISO 14001 and SA 8000 management standard with exports. Thirdly, ranking approach was used of top sixteen ISO 9001, ISO 14001 and SA 8000 certified countries globally, on the basis of weights provided by GRA models. Further, decision parameters were set up according to alternatives. Finally conservative model was utilized to understand which certification have more influence on company exports among top sixteen developing and developed countries.

GRA models, also known as a Grey Incidence Analysis (GIA) model, is commonly considered as a preliminary component of the grey system theory Badri Ahmadi et al. [53]. Grey system theory has its underlying foundations in the work done by Deng in 1982 [54]. From that point forward it has been utilized in different fields of learning to manage the issues of vulnerability in a system with poor information and obscure data sources Ikram et al. [55]. A significant advantage of using this method is its applicability in the case of a small sample size Dejamkhooy et al. [54].

In 1985, Deng [56] first introduced the idea of grey relational grade (GRG) in GRA model. As indicated by Wang et al. [57], the grey incidence model is considered to be superior over statistical models for analyzing system dynamics. The main advantage of the grey incidence model is solving the uncertainty of system issues whose physical mechanisms and operating prototype lack complete physical information (Javed et al. [58]). The fundamental idea of GIA is to utilize

the level of closeness of the geometric curve of accessible information to decide if their associations are close or not Liu et al. [59]. Consequently, it can be noted that this model is a benchmark for calculating the extent of the relationship between variables. According to eng and Shen [60] the high level of similarity indicates a closer relationship between intervals or sequences.

3.2.1. GRA model of Deng's degree

Several GIA models have been introduced by scholars, yet the model proposed by Deng [56] is more effective Bayramoglu and Hamzacebi [61] and plays a key role in the decision making methodology of grey system theory as noted by Goyal and Grover [62]. Taking this into account, real life issues in numerous fields associated with logical or scientific reasoning have utilized this approach Zuo et al. [63]. According to Javed and Liu [64] the steps for calculating Deng's grey incidence of two sets of data indicated by X_0 and X_1 are as follows:

- **Stage 1:** First step involved the computation of initial image of X_0 and X_i , $i = 1, 2, \dots, m$, where:

$$X'_i = X_i/x_i(1) = (x'_i(1), x'_i(2), \dots, x'_i(n)); i = 0, 1, 2, 3 \dots, m. \quad (1)$$

- **Stage 2:** Compute the difference of sequence X'_0 and X'_i $i = 1, 2, 3, \dots, m$, as:

$$\Delta i(l) = \|x'_0(l) - x'_i(l)\|, \Delta = (\Delta i(1), \Delta i(2), \dots, \Delta i(n)), i = 1, 2, 3, \dots, m. \quad (2)$$

- **Stage 3:** Step three involves the findings of maximum and minimum differences i.e.

$$\begin{aligned} M &= \max_i \max_l \Delta_i(L), \\ m &= \min_i \min_l \Delta_i(L). \end{aligned} \quad (3)$$

- **Stage 4:** In the fourth step we estimate the coefficients of grey incidence (also referred as, Grey/Point relation coefficients) by using the following formulae:

$$\gamma_{0i}(l) = \frac{m + \xi M}{\Delta i(l) + \xi M}; \xi \in (0, 1); l = 1, 2, \dots, n; i = 1, 2, \dots, m, \quad (4)$$

Whereas, ξ denotes the distinctive coefficient. We suppose the value is $\xi = 0.5$ as a distinctive coefficient Javed and Liu [64].

- **Stage 5:** The last step is to compute the grey relational degree (GRG) also known as Deng's degree of grey incidence by:

$$\gamma_{0i} = \frac{1}{n} \sum_{l=1}^n (\gamma_{0i}(L)); i = 1, 2, \dots, m. \quad (5)$$

We change the fractional form $1/n$ with weights w_k , as

$$\gamma_{0i} = \sum L = 1n(\gamma_{0i}(L) \times w_l); i = 1, 2, 3 \dots, m. \quad (6)$$

If the impact of each factor on the framework is not the same, the sum of weights will be equal to one, i.e. $\sum w_l = 1$ as obtained from Sarikaya and Güllü [65].

Basically, $\frac{1}{n}$ suggests equally distributed weighting criteria whereas w_k is an unequally distributed weighting scheme representing real life, or complex challenges. So, the first half of Deng's GRA model can be connected as every single free factor is similarly weighted in the top fifteen certified ISO9001, SA8000 and ISO14001 countries.

3.2.2. Absolute degree of GIA model

Research conducted by [66] concluded that Deng's GIA reveals insight into impacts among the components (represented by the grey data sequence). Whereas absolute degree grey incidence analysis (ADGIA) models, also known as absolute GRA models, reveals insight into the relationships among those factors Zhicai and Li [67]. Use of the two models means to accomplish the goal of finding relationships among factors by assessing the geometric vicinity between the sequence curves and has substantial worth in the literature Javed and Liu [68]. Below are the steps necessary to evaluate the absolute degree of Grey Incidence at two equally interval of time sequences X_0 and X_1 .

- **Stage 1:** The involves the estimation of the zero-starting point images X_0^0 and X_1^0 based on the interval sequence in X_0 and X_1 .
- **Stage 2:** Second, compute $|s_0|$, $|s_1|$ and $|s_1 - s_0|$.
- **Stage 3:** To compute the absolute degree of grey incidence represent by (ϵ_{01}) among the time interval sequence X_0 and X_1 in the last step by using the following formulae:

$$\epsilon_{01} = \frac{1 + |s_0| + |s_1|}{1 + |s_0| + |s_1| + |s_1 - s_0|}. \quad (7)$$

For a more detailed explanation on the two GIA models, Deng's GRA and absolute GRA and their characteristics, see Liu et al. [59].

3.2.3. Deng degree of GRA model

Deng's GRA model proposes using the Deng degree of grey incidence or GRG also known as Deng's grey incidence analysis (DGIA) model. It is operationalized as

$$\gamma_{0i} = \sum_{l=1}^n (\gamma_{0i}(L) \times w_l); i = 1, 2, 3, \dots, m, \quad (8)$$

or

$$\gamma_{0i} = \frac{1}{n} \sum_{l=1}^n (\gamma_{0i}(L)); i = 1, 2, 3, \dots, m, \quad (9)$$

Where $\gamma_{0i}(L)$ represents coefficient of grey incidence or coefficient of grey relational (GRC). The first form of DGIA represents the weighted form of GRA/DGIA and second represents as non-weighting scheme (or equally weights).

3.2.4. Absolute Degree Grey Incidence Analysis (ADGIA)

Absolute GRA model introduced the absolute degree of grey relation also known as ADGIA model. We employed ADGIA to calculate the weights of top ISO 9001, 14001 and SA8000 certified companies in developing and developed countries. It is written as:

$$\varepsilon_{01} = \frac{1 + |s_0| + |s_1|}{1 + |s_0| + |s_1| + |s_1 - s_0|} \quad (10)$$

3.2.5. Zero-scale point image

Let the data sequence of system behavior represent as $X_i = (x_i(1), x_i(2), \dots, x_i(n))$ and C indicate the operator of sequence, which satisfies and $X_i(L)c = (x_i(L)c - x_i(1)); L = 1, 2, 3, \dots, n$. Then, C is a point operator starting from zero and X_iC known as zero-scale point image of X_i . X_iC is often written

$$X_iC = X_i^0 = (x_i^0(1), x_i^0(2), \dots, x_i^0(n)) \quad (11)$$

According to Liu et al. [69], we suppose that the length of variables X_i and X_j are same of one-time-interval sequences and X_i and X_j are zero-starting point images follow as :

$$X_i^0 = (x_i^0(1), x_i^0(2), x_i^0(3), \dots, x_i^0(n)), \quad (12)$$

$$X_j^0 = (x_j^0(1), x_j^0(2), x_j^0(3), \dots, x_j^0(n)). \quad (13)$$

Then

$$|s_i| = \left| \sum_{l=2}^{n-1} x_i^0(L) + \frac{1}{2} x_i^0(n) \right|, \quad (14)$$

$$|s_j| = \left| \sum_{l=2}^{n-1} x_j^0(L) + \frac{1}{2} x_j^0(n) \right|, \quad (15)$$

$$|s_i - s_j| = \left| \sum_{l=2}^{n-1} (x_i^0(L) - x_j^0(L)) + \frac{1}{2} (x_i^0(n) - x_j^0(n)) \right| \quad |s_j| = \left| \sum_{l=2}^{n-1} x_j^0(L) + \frac{1}{2} x_j^0(n) \right| \quad (16)$$

3.2.6. The Second Synthetic Degree of Grey Incidence Analysis (SSDGIA) model

The SSDGIA model proposed by Javed and Liu [64], or the Second Synthetic GRA model, is given by:

$$\rho_{ij} = \theta \varepsilon_{ij} + (1-\theta) \gamma_{ij}; \theta \in [0, 1], \quad (17)$$

Where ρ_{ij} denotes the SS degree of grey incidence/grey relation (SSGRG), ε_{ij} denotes absolute degree of grey incidence/grey relation (absolute GRG) and γ_{ij} represents. Deng's degree of grey incidence/grey relation (GRG). Deng's GRA model depends on grey relational/incidence coefficient of certain points; however, absolute GRA model depends on integral (relatively comprehensive) perspective Liu et al. [69]. The SSDGIA considers overall intimacy between two levels of sequences and an integral position. The first synthetic degree grey incidence analysis (SDGIA) model was introduced in accordance with the absolute and the relative degree of grey incidence, and can be now be found Javed and Liu [64].

The mechanism of SSDGIA model has been created based on the working standard of SDGIA model. It is important to consider $\theta = 0.5$ while making decisions in order to accomplish a thorough positioning fusing the advantages of the two terms i.e. γ and ε , without favoring one over another and if it is essential, then θ can alter. If you wish to support γ , at that point estimation of θ can be diminished and if you need to favor ε , at that point θ this can also be supported.

Table 1
Grey relational evaluation of ISO 9001 for developed and developing countries.

Country/ISO9001	GRG	Ranking	Absolute GRG	Ranking	SS GRG	Ranking
China	0.620963	6	0.949369	10	0.785166	6
Italy	0.626138	5	0.970589	5	0.798363	5
Japan	0.623523	4	0.992179	2	0.807851	4
Spain	0.674034	2	0.970628	4	0.822331	2
Germany	0.461477	14	0.858369	13	0.659923	13
USA	0.542428	12	0.683683	14	0.613056	14
India	0.567609	10	0.995446	1	0.781528	7
France	0.500898	13	0.884656	12	0.692777	12
Brazil	0.554957	11	0.93582	11	0.76532	11
UK	0.457467	15	0.56313	15	0.51029	15
Romania	0.60604	7	0.95341	9	0.77973	8
Pakistan	0.58363	8	0.95931	8	0.77147	10
Bulgaria	0.57902	9	0.96898	7	0.77411	9
Vietnam	0.67159	3	0.96949	6	0.82054	3
Portugal	0.68976	1	0.98641	3	0.83803	1

4. Result and discussion

All three grey relation models were successfully operationalized. The analysis of data of the ISO 9001, ISO 14001 and SA 8000 top certified countries are shown in [Tables 1–3](#). According to Liu et al. [70] the relationship between data sequence in grey incidence and degree of grey incidence models cannot be zero, thus their insignificance values can fluctuate. [Table 1](#) shows the value of GRG, absolute GRG and SSGRG models of fifteen ISO 9001 top certified countries. For instance, in the case of the GRG model, Portugal's exports indicate a strong association with quality management standard by obtaining the highest weight (0.68976) followed by Spain and Vietnam by obtaining weights (0.674034) and (0.67159) respectively. Whereas, Germany and the UK considered the least influencing association between adoption of ISO 9001 certification and export of good and services by obtaining weights (0.461477) and (0.457467) respectively. This supports the idea of adopting uniform standards, like ISO 9001 that are acceptable for numerous clients, as these standards can enable reduced transaction costs [71]. These benefits are likely going to be greater for firms whose engagement in foreign markets is bigger and, specifically, for those who trade many products to numerous nations. In the case of the UK, the adoption of ISO 9001 indicates a weak relationship with country exports of products and services. From prior work we know there is a positive connection between appraisals of ISO 9001 accreditation costs for each unit of sales and overall sale Cândido et al. [72].

In developing countries like China, India, Pakistan, Brazil and Vietnam, adoption of ISO 9001 certification shows a stronger association with the export of goods and services than the developed countries presented in [Figs. 1 and 2](#). Destination countries are subsequently a vital factor for clarifying why firms may look for accreditation. Thus, firms whose exports are primarily located in countries with elevated amounts of accreditation activities should have more motivation to acquire ISO accreditation Javorcik and Sawada [73].

Similarly, in the case of absolute GRG, Indian exports show a more positive association with adoption of quality management standards over all other ISO 9001 accredited countries. A weak relationships were observed in the case of the UK. Italy, Brazil, the UK, and Pakistan exhibit the same ranking on the basis of GRG and absolute GRG. In developing countries like China, India, Pakistan, Brazil and Vietnam, adoption of ISO 9001 certification shows an average value of absolute GRG (0.961887) and in developed countries like Italy, Japan, Spain, Germany, the USA, France, the UK, Romania, Bulgaria and Portugal, the average value was 0.883203. This implies that in developing countries, adoption of ISO 9001 contributes more significantly to economic development than in developed countries. Overall, the SSGRG model shows the export is most significantly associated with ISO 9001 in Portugal and least significantly in the UK.

[Table 2](#) shows the value of GRG, absolute GRG and SSGRG models of the top fifteen ISO 14001 certificated countries. For instance, the Deng GRG model indicated that ISO certified companies in India have strong exports by obtaining the highest weight (0.705312), however, the Spanish economy showed weak association towards registration of the ISO 14001 standard by obtaining weight (0.53363). In developing countries like China, India, Pakistan, Brazil and Vietnam, adoption of the ISO 14001 certification shows a stronger association with the export of goods and services than developed countries.

The overall significance of clients requiring certification is likely different from sector to sector. Generally, progressive organizations demand that their suppliers should be ISO certified Wiengarten et al. [74]. There are a few explanations that describe what we have found in the data and these relationships. Accreditation decreases the immediate expense of forthcoming inspections faced by buyers, the costs related to compliance, and the joint cost of contracting. Basically, organizations selling to different producers are bound to be certified with ISO 9001.

From the Deng GRG model's interesting results, we see the absolute GRG models indicated a positive and significant relationship between economic development and ISO 14001 adoption. For example, Italy shows the weakest association and obtained the lowest weight (0.847028). Environmental groups, investors and locals demanding sound corporate environmental management systems can acknowledge ISO 14000 certification as proof of an organization's commitment

Table 2

Grey relational evaluation of ISO 14001 for developing and developed countries.

ISO14001	Deng GRG	Rank	Absolute GRG	Rank	SS GRG	Rank
China	0.667075	4	0.953042	4	0.808222	4
Japan	0.639624	7	0.912077	8	0.77585	7
Italy	0.581637	12	0.847028	15	0.714333	14
Spain	0.53363	15	0.890153	10	0.711891	15
UK	0.566296	13	0.867572	14	0.716934	13
Germany	0.565526	14	0.936011	7	0.750768	9
USA	0.640939	6	0.972095	3	0.806517	5
France	0.588172	11	0.880285	11	0.734228	12
Romania	0.606282	9	0.896479	9	0.751381	8
India	0.705312	1	0.978593	2	0.841953	2
Brazil	0.69678	2	0.99178	1	0.84428	1
Pakistan	0.63271	8	0.86813	13	0.75042	10
Bulgaria	0.66457	5	0.93741	6	0.80099	6
Vietnam	0.66907	3	0.95212	5	0.8106	3
Portugal	0.59297	10	0.87915	12	0.73606	11

to reducing ecological problems. Certification under this standard moderates the concerns of partners and enables firms to connect to new export markets, and to overcome ISO 14000 as a trade barrier Murmura et al. [75].

Overall, the SSGRG model shows similar results as presented by GRG model in Table 2. In developing countries like China, India, Pakistan, Brazil and Vietnam, adoption of ISO 14001 certification shows an average value of SSRGR (0.811095) and in developed countries like Italy, Japan, Spain, Germany, the USA, France, the UK, Romania, Bulgaria and Portugal, the average SSGRG value was (0.749895), which implies that adoption of ISO 140001 contributed more to the economic development of developing countries than developed countries (see Fig. 3).

In developed countries like the USA, respondents in general tend to view the standard with doubt, and the number of ISO 14001 accreditations in the USA remains low when compared to other developed countries Hikichi et al. [76]. In spite of this fact, the low ISO 14001 certification is likely connected to the USA's environmental regulatory culture and its dependence on a complex compliance system Neves et al. [77]. The firms surveyed in the USA frequently showed up very concerned and involved in complying with domestic environmental commitments and U.S. EPA regulation that minimal corporate energy for voluntary activities, such as ISO 14001 Ikram et al. [55].

In developing countries, the adoption of ISO 9001, ISO 14001 and SA 8000 management standards contributed to the countries' economic development comparatively more so than in developed countries. This is likely due to the increased pressures from stakeholders and the increase in export volume shown in Fig. 2. The data analysis by GRG, absolute GRG and SSGRG is in Table 3 of the top SA 8000 certified countries. Interestingly, Pakistan's exports increased due to compliance of SA8000 based on the Deng GRG and SSGRG models by obtaining the weights (0.88484) and (0.874359) respectively. This positive association is a signal for the Pakistan economy towards more sustainable development through registration of SA 8000 standard. The weakest relationship was observed in the case of Spain. According to absolute GRG model, China obtained (0.992269) weight, showed the strongest association and the UK indicated the weakest relationship among SA8000 accreditation and the export of goods and services by obtaining the weight (0.62649). The adoption of the SA8000 social standard is new and more nuanced tool to enhance the trade and competitive advantage of China's international trade. The perceived benefits of SA8000 certification include overcoming social issues, knowing the value of enterprise, setting up emergency realizations, short term foreign investment with the aim to manage this new issue, and additionally support their own management Prakash and Potoski [78]. For the benefit of economic development, the government should be aware of relevant laws and regulations, improve and impose them, and this holds true for labor laws.

The relational evaluation of SA8000 by SSGRG models also shows the same results for Pakistan, the strongest relationship as indicted by GRG model. Economic development in Spain was observed to be influenced the least by SA8000 registration out of all fifteen selected countries in the SSGRG model. Overall, the results show that the developing countries' export economy was increased when sustainability standards such as ISO 9001, ISO 14001 and SA 8000 are in place as this is supported by three novel grey relational models.

Deng GRA and absolute GRA models are showing that among ISO 9001, ISO 14001 and SA8000 management standards, the export of good and services by top certified countries have shown to be the most strongly associated with the effective implementation of these three management standards. The second synthetic GRA model reveals the reliability of the top fifteen certified quality, environmental and ethical standards to be important factors contributing to a country's economic development as shown in descending order in Table 3.

Based on the data analyzed by the Second Synthetic Degree of Grey Incidence Analysis (SSDGIA) model in the selected developing and developed countries, ranking criteria was established on adoption of ISO 9001, ISO 14001 and SA8000 certification in Table 4. China's economy showed very significant effect due to ISO 14001 than ISO 9001 and SA8000 in order to make organizations more sustainable. We can conclude from the results that environmental regulations in China contributed significantly to the development of the international trade sector. Despite economic development as an important factor, environmental regulation also stimulates exports and imports. An important finding

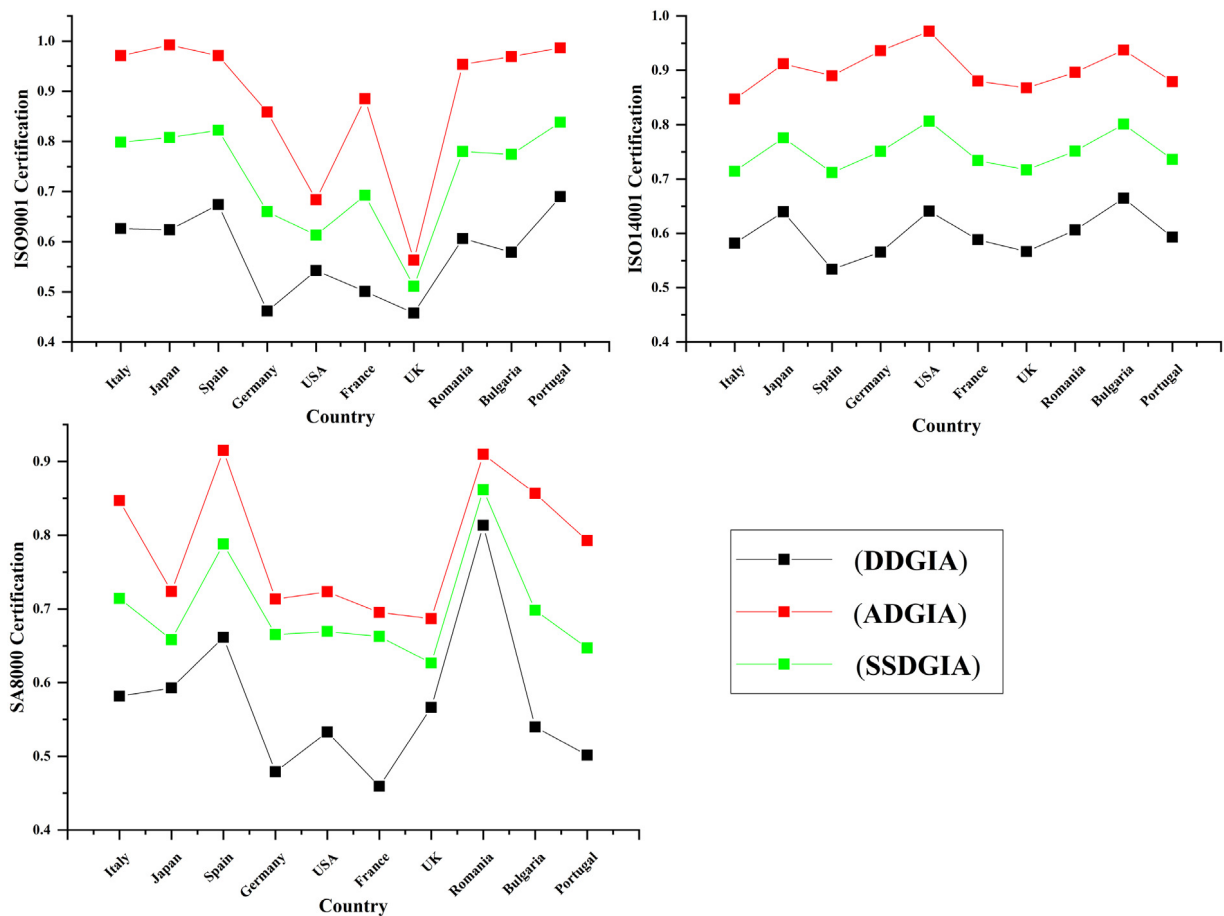


Fig. 1. ISO 9001, ISO 14001 and SA8000 comparative grey relational evaluation of developed countries.

Table 3

Grey relational evaluation of SA8000 for developing and developed countries.

SA8000	Deng GRG	Rank	Absolute GRG	Rank	SS GRG	Rank
Italy	0.581637	9	0.847028	9	0.714333	8
India	0.742548	3	0.948966	4	0.845757	5
China	0.721099	4	0.992269	1	0.856684	3
Romania	0.813392	2	0.90958	6	0.861486	2
Pakistan	0.88484	1	0.863878	7	0.874359	1
Brazil	0.713615	5	0.982862	2	0.848238	4
Bulgaria	0.539628	11	0.856708	8	0.698168	9
Viet Nam	0.622929	7	0.970305	3	0.796617	6
Spain	0.661371	6	0.914853	5	0.788112	7
Portugal	0.501412	13	0.792557	10	0.646985	14
Japan	0.59297	8	0.72354	11	0.65825	13
Germany	0.47885	14	0.71325	13	0.66521	11
USA	0.53291	12	0.72314	12	0.66951	10
France	0.45926	15	0.69521	14	0.66254	12
UK	0.56626	10	0.68672	15	0.62649	15

is that environmental regulations are advantageous for improving Chinese international trade. Following adoption of environmental regulations, the net exports of primary products, which are resource intensive and contribute to pollution, have decreased, while green value-added manufactured products increased He and Shen [79].

According to ISO, China achieved the highest number of ISO 14001 certifications of any other country. Moreover, adoption of ISO 9001 displayed a higher contribution to the economic development of Italy from 2002 to 2017. Meanwhile, due to the adoption of ISO 9001 and SA8000, the UK economy was observed to have an insignificant relationship to the exports of goods and services. The same impact of the certification of ISO 9001 and ISO 14001 on economic growth was

Table 4
Ranking of QES dimension contribution to export of goods and services.

Certification	Models	Ranking
ISO9001	GRG	Portugal > Spain > Vietnam > Japan > Italy > China > Romania > Pakistan > Bulgaria > India > Brazil > USA > France > Germany > UK
	Absolute GRG	India > Japan > Portugal > Spain > Italy > Vietnam > Bulgaria > Pakistan > Romania > China > Brazil > France > Germany > USA > UK
	SS GRG	Portugal > Spain > Vietnam > Japan > Italy > China > India > Romania > Bulgaria > Pakistan > Brazil > France > Germany > USA > UK
ISO14001	GRG	India > Brazil > Vietnam > China > Bulgaria > USA > Japan > Pakistan > Romania > Portugal > France > Italy > UK > Germany > Spain
	Absolute GRG	Brazil > India > USA > China > Vietnam > Bulgaria > Germany > Japan > Romania > Spain > France > Portugal > Pakistan > UK > Italy
	SS GRG	Brazil > India > Vietnam > China > USA > Bulgaria > Japan > Romania > Germany > Pakistan > Portugal > France > UK > Italy > Spain
SA8000	GRG	Pakistan > Romania > India > China > Brazil > Spain > Vietnam > Japan > Italy > UK > Bulgaria > USA > Portugal > Germany > France
	Absolute GRG	China > Brazil > Vietnam > India > Spain > Romania > Pakistan > Bulgaria > Italy > Portugal > Japan > USA > Germany > France > UK
	SS GRG	Pakistan > Romania > China > Brazil > India > Vietnam > Spain > Italy > Bulgaria > USA > Germany > France > Japan > Portugal > UK

seen in the case of Romania, presented in Table 4. Here it is noticed that when Deng's GIA and Absolute GIA models produce opposite outcomes, the SSDGIA model (at $\theta = 0.5$) showed moderate outcomes. In the present study, as long as the grey impact is concerned with the economic development (exports of goods and services) of developed and developing countries, quality management standard, ISO 9001 and SA 8000 remain in the same position, while ISO 14001 standard switched its position. As far as Grey association is concerned, no consistency exists among the orders. Likewise, as per GIA analysis, there is no undeniable consistency in the prioritization of the factors. In this manner the present study contends that the outcome concerning prioritization of quality measurements acquired was bound to be a coincidence and subsequently it is difficult to summarize.

The decision analysis helped to quantify the difference between the export of goods and services between all the top ISO 9001, ISO 14001 and SA 8000 certified countries. So, for this purpose, a first step, the states of nature and alternative actions, were characterized and appeared in Table 5. Where, $m = 3$, $n = 15$, and outcome = $v(a_i, s_j)$ whereas $i = 1, 2$ and $j = 1, 2, 3, 4, 5, \dots, 15$. Let a_1, a_2 and a_3 represents the superiority analysis of adoption of ISO 9001, ISO 14001 and SA8000 on growth in export of good and services in US \$ in developed and developing countries. The S_1 to S_{15} represents the top fifteen ISO 9001, ISO 14001 and SA8000 certified countries globally. Table 6 represents the Second Synthetic (SS) Degree of Grey Incidence-based Matrix, reporting the "second synthetic grey relations" between the decision criteria and decision actions. Finally, we operationalized a decision making model to see the comparative analysis among ISO 9001, ISO 14001 and SA 8000 in developed and developing countries.

The Conservative (maximin) model

We next used, the following methodology proposed by Prasad [80]. As we maximized the function for international trade, so it is a maximin criterion:

$$\max a_i \{ \min s_j v(a_i, s_j) \} = \max a_i \left\{ \begin{matrix} 0.51029 \\ 0.711891 \\ 0.62649 \end{matrix} \right\} = 0.711891. \quad (18)$$

Our findings demonstrate that economic development is expected to be positively impacted by the adoption of ISO 14001 standard in developing and developed nations. The outcomes of the decision making approach show that environmental management systems are expected to play a significant role in developing nations than in developed nations. In this way, there is a legitimate need to enhance management systems to include economic and environmental sustainability.

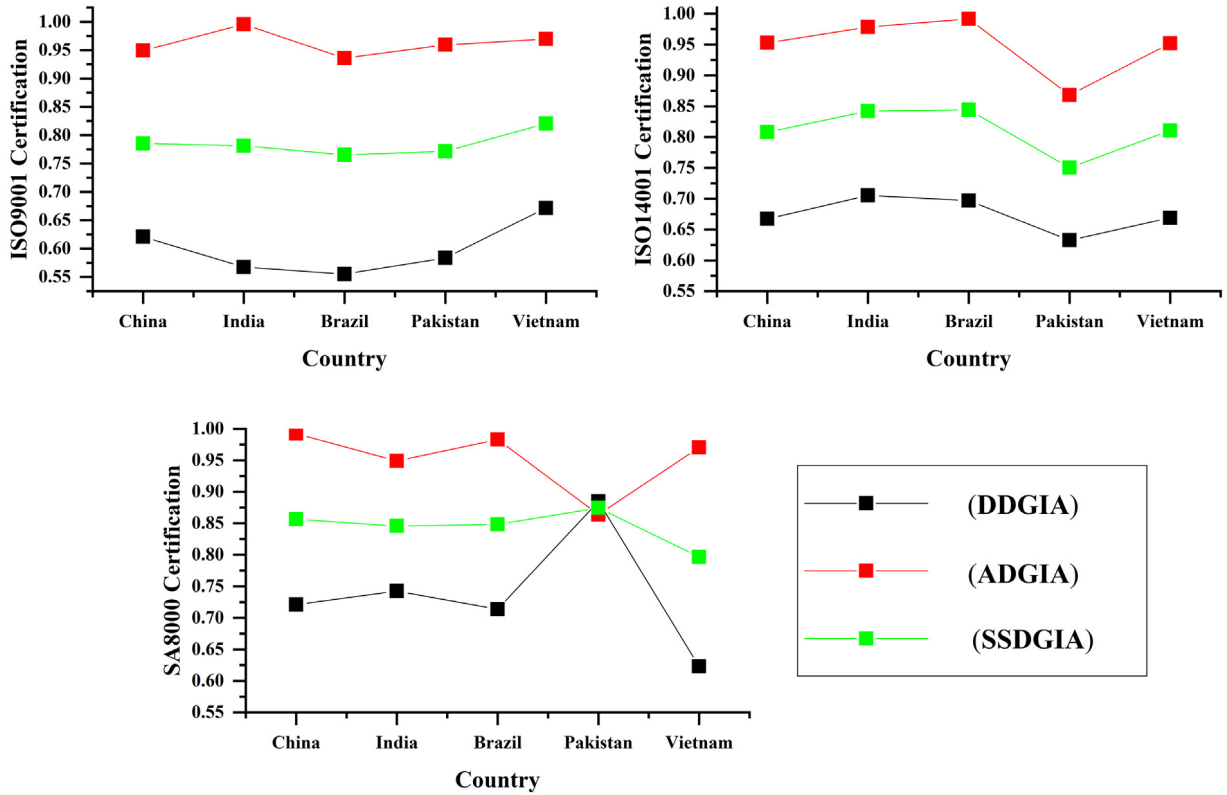


Fig. 2. ISO 9001, ISO 14001 and SA8000 comparative grey relational evaluation of developing countries.

Table 5

Summary of decision parameters.

Goal	Measuring grey relation (Association) between ISO9001, ISO14001, SA8000, and Export of Product & Services of developing and developed countries
Criteria's(S_j) $j = 1, 2, 3, \dots, n$	China (S_1) Italy(S_2) Japan(S_3) Spain(S_4) Germany(S_5) USA(S_6) India(S_7) France(S_8) Brazil(S_9) UK(S_{10}) Romania(S_{11}) Pakistan(S_{12}) Bulgaria(S_{13}) Viet Nam(S_{14}) Portugal(S_{15})
Alternatives (a_i); $i = 1, 2, \dots, m$	Grey relation of ISO 9001 was superior in developed vs developing countries (a_1) Grey relation of ISO14001 was superior in developed vs developing countries (a_2) Grey relation of SA8000 was superior in developed vs developing countries (a_3)

5. Conclusion

Contributions of this study fill a gap in the limited literature in regard to the impact of international standards on international trade in both developing and developed nations. According to best of our knowledge this study is the first

Table 6
Criteria action matrix.

SS GRG	a_1	a_2	a_3
S_1	0.785166	0.808222	0.856684
S_2	0.798363	0.714333	0.714333
S_3	0.807851	0.77585	0.65825
S_4	0.822331	0.711891	0.788112
S_5	0.659923	0.750768	0.66521
S_6	0.613056	0.806517	0.66951
S_7	0.781528	0.841953	0.845757
S_8	0.692777	0.734228	0.66254
S_9	0.76532	0.84428	0.848238
S_{10}	0.51029	0.716934	0.62649
S_{11}	0.77973	0.751381	0.861486
S_{12}	0.77147	0.75042	0.874359
S_{13}	0.77411	0.80099	0.698168
S_{14}	0.82054	0.8106	0.796617
S_{15}	0.83803	0.73606	0.646985

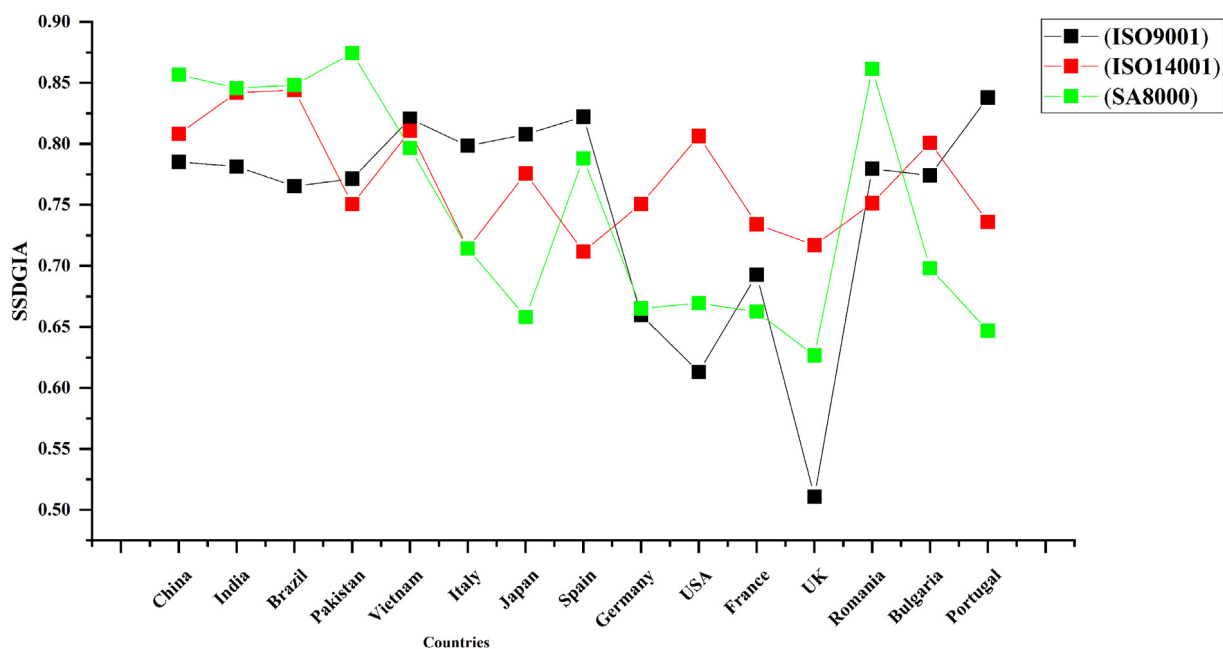


Fig. 3. Comparative analysis of QES of developing and developed countries based on SSDGIA.

of its kind that empirically investigate and discuss in detail whether the adoption of ISO 9001, ISO 14001 and SA8000 quality standards will improve an organization's exports. Contributions of this study also include new insights from our comparative analysis among the top certified developing and developed countries. The aim of this study is to fill in the gap in the literature by synthesizing the impacts of ISO 9001, ISO 14001 and SA8000 certification on the exports of developed and developing nations at a country level of analysis. To address endogeneity issues in acquiring the standards, we utilized a novel mathematic model, grey relational models, to both analyze and compare the average impact of accreditation among treated and control groups. Our results demonstrate that getting ISO 9001, ISO 14001 and SA 8000 accreditation fundamentally increases the export of goods and services and there are a number of other benefits that go with these increased exports.

Certification with ISO 9001 can help firms to conquer information barriers when it comes to entering new export markets or increasing their activities internationally as well as take into account internal efficiency gains. These consequently lead to improved outcomes in the form of exports. In developed nations, for example, Portugal and Spain, exports were substantially improved because of ISO 9001 certification. This signals, quality standards as offering a 'minimal cost' of an organization's commitment to quality. The USA and UK demonstrated a feeble and negative association with the adoption of ISO 9001 quality standards on national exports of goods and services. ISO 14001 certification for developing countries is an essential product characteristic that needs further investigation. There are critical geographical insights in developing nations. For example, China clearly leads the world in ISO 14001 standard use and certification. The decision

makers in Asian nations have adopted these voluntary activities and stakeholders have demanded the enhancement of the industry's ecological performance in the region. This makes for a dynamic environment where EMS accreditation can bring about additional accreditation in supply chains. The insights from our findings reveal that European clients are commonly more environmentally conscious than customers in other developed regions of the world that fundamentally impacts the exports in developing nations.

It is noteworthy that developing nations' exports indicate an ascending pattern regarding compliance with the SA8000 standard and the GRG model. This positive relationship is a signal for developing country economies towards sustainable development through the use of the SA 8000 standard certification.

Our findings have implications for both export business managers and policymakers. Export business managers need to understand the significant task of adoption of important standards and execute programs supporting quality, environmentally friendly and socially sustainable business practices as they can provide advantages for product differentiation, business performance as well as international trade. Governments and policymakers should attempt enable current and future exporters the use of quality, ecological and social standards to improve export business as this will help enterprises improve market presence worldwide and enhance export performance.

To further improve both policy and decision analysis, there is an opportunity to develop education systems with courses concentrating on ecological, quality and social issue. These evidence based management can leverage best practices of exporters and their quality, environmentally friendly, and socially responsible practices. Governments can develop initiatives to integrate and publicize the benefits at a country level concerning global standard prerequisites, highlight special projects, and provide incentives to firms who may be skeptical of embracing environmentally friendly activities in developed and developing nations.

Finally, taking into account the challenging task of diminishing social, ecological and economic issues, and making the organization sustainable, this study suggests the adoption of ISO 9001, ISO 14001 and SA8000 accreditation is both beneficial and a move in the right direction for more sustainable business practices. More research needs to be done in this area involving multimethod studies, modeling and testing of relationships to help determine the important attributes of successful firms and countries improving the quality, environmental and social performance of their supply chains and exported products.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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